

Integration of Trigo. Functions

Here are some rules of Integration of Trigonometric Functions:

	Differentiation 微分	**Integration 積分**
1.	$\frac{d}{dx}(-\cos x) = \sin x$	$\int \sin x \, dx = -\cos x + C$
2.	$\frac{d}{dx}(\sin x) = \cos x$	$\int \cos x \, dx = \sin x + C$
3.	$\frac{d}{dx}(\tan x) = \sec^2 x$	$\int \sec^2 x \, dx = \tan x + C$
4.	$\frac{d}{dx}(-\cot x) = \csc^2 x$	$\int \csc^2 x \, dx = -\cot x + C$
5.	$\frac{d}{dx}(\sec x) = \sec x \tan x$	$\int \sec x \tan x \, dx = \sec x + C$
6.	$\frac{d}{dx}(-\csc x) = \csc x \cot x$	$\int \csc x \cot x \, dx = -\csc x + C$

where C is integration constant

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Integration of Basic Trigo.

e.g. ***1.*** $\int (\sin x + \cos x)\, dx = -\cos x + \sin x + C$



2. $\int (\sin x - \cos x)\, dx =$

3. $\int (\sec^2 x + \csc^2 x)\, dx =$

4. $\int \dfrac{dx}{\cos^2 x} =$

5. $\int \dfrac{\sin^2 x + 1}{\sin^2 x}\, dx =$

6. $\int (\tan x + \sec x)\sec x\, dx =$

7. $\int \dfrac{\sin x}{\cos^2 x}\, dx =$

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Integration of Trigo. with compounded angle

計算涉及**複角 (Compounded Angles)** 的三角函數的積分時，我們會運用**代換積分法 (Integration by substitution)**：

e.g. Find $\int \sin 2x\, dx$.

Let $u = 2x$

Then $\dfrac{du}{dx} = 2$, i.e. $du = 2dx$

$$\therefore \int \sin 2x\, dx = \int \frac{1}{2}\sin 2x \cdot 2dx$$

將 dx 寫成 $\dfrac{1}{2}\cdot 2dx$

$$= \frac{1}{2}\int \sin u\, du$$

$$= \frac{1}{2}(-\cos u) + C$$

$$= -\frac{1}{2}\cos 2x + C$$

我們亦可以直接用以下兩條公式，極速計算涉及一些複角的三角函數的積分：

$$\int \sin(ax+b)\, dx = -\frac{1}{a}\cdot\cos(ax+b) + C$$

$$\int \cos(ax+b)\, dx = \frac{1}{a}\cdot\sin(ax+b) + C$$

Let's TRY IT!

[A.Math 2004 Q.1(a)]

Find $\int \cos(3x+1)\, dx$. ***(2 marks)***

Integration of product of sinx and cosx

由於積分沒有積法則 (NO Product Rule!)，因此計算三角函數相乘的積分時，我們會運用恒等式 (Identities) 將相乘轉換成相加 (Product to Sum)：

- $\sin A\cos B = \frac{1}{2}[\sin(A+B) + \sin(A-B)]$
- $\cos A\sin B = \frac{1}{2}[\sin(A+B) - \sin(A-B)]$
- $\cos A\cos B = \frac{1}{2}[\cos(A+B) + \cos(A-B)]$
- $\sin A\sin B = -\frac{1}{2}[\cos(A+B) - \cos(A-B)]$

e.g. Find $\int \sin 2x\cos 3x\,dx$.

$$\int \sin 2x\cos 3x\,dx$$
$$= \int \frac{1}{2}[\sin(2x+3x)+\sin(2x-3x)]\,dx$$
$$= \frac{1}{2}\int [\sin 5x - \sin x]\,dx$$
$$= \frac{1}{2}\left(\frac{-\cos 5x}{5} + \cos x\right) + C$$
$$= -\frac{1}{10}\cos 5x + \frac{1}{2}\cos x + C$$

Product to Sum
相乘轉換成相加

[A.Math 2009 Q.1(b)]

Find $\int \sin 3\theta\cos\theta\,d\theta$. ***(3 marks)***

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Integration of $\sin^2 x$ and $\cos^2 x$

由於積分沒有鏈式法則 (NO Chain Rule)，因此計算 $\sin^2 x$ 及 $\cos^2 x$ 的積分時，我們會運用恒等式將它們轉換成可被積分的形式：

$\cos 2x = 1 - 2\sin^2 x$

$\cos 2x = 2\cos^2 x - 1$

$$\sin^2 x = \frac{1}{2}(1 - \cos 2x)$$

$$\cos^2 x = \frac{1}{2}(1 + \cos 2x)$$

e.g. Find $\int \sin^2 x \, dx$.

$$\int \sin^2 x \, dx$$
$$= \int \frac{1}{2}(1 - \cos 2x) \, dx$$
$$= \frac{1}{2}\int (1 - \cos 2x) \, dx$$
$$= \frac{1}{2}\left(x - \frac{\sin 2x}{2} + C\right)$$
$$= \frac{1}{2}x - \frac{1}{4}\sin 2x + C$$

OR

$$\int \sin^2 x \, dx$$
$$= \int (\sin x \cdot \sin x) \, dx$$
$$= \int -\frac{1}{2}[\cos(x + x) - \cos(x - x)] \, dx$$
$$= -\frac{1}{2}\int (\cos 2x - 1) \, dx$$
$$= -\frac{1}{4}\sin 2x + \frac{1}{2}x + C$$

Let's TRY IT!

[A.Math 2003 Q.1]

Find $\int \cos^2 \theta \, d\theta$. ***(3 marks)***

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Integration of $\sin^n x$ and $\cos^n x$ (n is odd)

當要計算 $\int \sin^n x\,dx$ ，其中 ***n*** **是單數**時，我們可用**代換積分法 (Integration by substitution)**，設 $u = \cos x$，並以三角恒等式輔助作積分：

e.g. Find $\int \sin^3 x\,dx$.

Let $u = \sin x$

Then $du = \cos x\,dx$

$\therefore \int \sin^3 x\,dx = \ldots$



Let $u = \cos x$

Then $du = -\sin x\,dx$

$$\therefore \int \sin^3 x\,dx$$

$$= \int -\sin^2 x \cdot (-\sin x)\,dx$$

$$= \int (\cos^2 x - 1) \cdot (-\sin x)\,dx$$

$$= \int (u^2 - 1)\,du$$

$$= \frac{u^3}{3} - u + C$$

$$= \frac{\cos^3 x}{3} - \cos x + C$$



[A.Math 1988 Paper II Q.8(a) Modified]

Using the substitution $y = \sin x$, find $\int \cos^7 x\,dx$. ***(5 marks)***

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Integration of $\sin^n x$ and $\cos^n x$ (n is even)

當要計算 $\int \sin^n x\,dx$ ，其中 ***n* 是雙數**時，我們則要用恒等式將算式中所有三角函數的**次方歸約 (Reducing Power)** 才能積分：

e.g. Find $\int \sin^4 x\,dx$.

Let $u = \cos x$

Then $du = -\sin x\,dx$

$\therefore \int \sin^4 x\,dx$

$= \int -\sin^3 x \cdot (-\sin x)\,dx$

$= -\int (\sin x)^3 \cdot (-\sin x)\,dx$

$= -\int \left(\sqrt{1-\cos^2 x}\right)^3 \cdot (-\sin x)\,dx$

$= -\int \left(1-u^2\right)^{\frac{3}{2}}\,du$

$= .....$



$\int \sin^4 x\,dx$

$= \int \left[\frac{1}{2}(1-\cos 2x)\right]^2 dx$

$= \frac{1}{4}\int (1-2\cos 2x+\cos^2 2x)\,dx$

$= \frac{1}{4}\int (1-2\cos 2x+\cos^2 2x)\,dx$

$= \frac{1}{4}\int [1-2\cos 2x+\frac{1}{2}(1-\cos 4x)]\,dx$

$= \frac{1}{4}\int (\frac{3}{2}-2\cos 2x-\frac{\cos 4x}{2})\,dx$

$= \frac{1}{4}\left(\frac{3}{2}x-\frac{2\sin 2x}{2}-\frac{\sin 4x}{8}\right)+C$

$= \frac{3}{8}x-\frac{1}{4}\sin 2x-\frac{1}{32}\sin 4x+C$

Let's TRY IT! Find $\int \cos^4 x\,dx$.

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Integration of product of ($\sin^m x \cdot \cos^n x$)

當要計算 $\int \sin^m x \cos^n x \, dx$ 時，我們會根據不同情況採用以下方法：

- ***m* 是單數 (*m* is odd)** ➔ 代換積分法 (Let $u = \cos x$)
- ***n* 是單數 (*n* is odd)** ➔ 代換積分法 (Let $u = \sin x$)
- **全是單數 (*m*, *n* are odd)** ➔ 代換積分法 (Let $u = \cos x$ or $\sin x$)
- **全是雙數 (*m*, *n* are even)** ➔ 恒等式次方歸約 (Reducing Power)

e.g. Find $\int \sin^3 x \cos^4 x \, dx$.

Let $u = \cos x$, then $du = -\sin x \, dx$

$$\begin{aligned}\int \sin^3 x \cos^4 x \, dx &= \int -\sin^2 x \cos^4 x \cdot (-\sin x) \, dx \\ &= \int (\cos^2 x - 1)\cos^4 x \cdot (-\sin x) \, dx \\ &= \int (u^2 - 1) u^4 \, du \\ &= \int (u^6 - u^4) \, du \\ &= \frac{u^7}{7} - \frac{u^5}{5} + C \\ &= \frac{\cos^7 x}{7} - \frac{\cos^5 x}{5} + C\end{aligned}$$



[A.Math 1981 Paper II Q.8(a) Modified]

Using the substitution $y = \sin x$, find $\int \cos^3 x \sin^2 x \, dx$. ***(6 marks)***

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Integration of $\tan^n x$

當要計算 $\int \tan^n x\,dx$ 時，我們會採用以下方法：

➢ ***n* 是單數 (*n* is odd)** ➔ 代換積分法 (Let $u = \sec x$)

有用恒等式： $\tan^2 x + 1 = \sec^2 x$

$$\tan^2 x = \sec^2 x - 1$$

e.g.1 Find $\int \tan x\,dx$

Let $u = \sec x$,
then $du = \sec x \tan x\,dx$

$$\begin{aligned}
&\therefore \int \tan x\,dx \\
&= \int \frac{1}{\sec x} \cdot \sec x \tan x\,dx \\
&= \int \frac{1}{u}\,du \\
&= \ln|u| + C \\
&= \ln|\sec x| + C
\end{aligned}$$

e.g.2 Find $\int \tan^3 x\,dx$

Let $u = \sec x$,
then $du = \sec x \tan x\,dx$

$$\begin{aligned}
&\therefore \int \tan x^3\,dx \\
&= \int \frac{\tan^2 x}{\sec x} \cdot \sec x \tan x\,dx \\
&= \int \frac{\sec^2 x - 1}{\sec x} \cdot \sec x \tan x\,dx \\
&= \int \left(u - \frac{1}{u}\right) du \\
&= \frac{u^2}{2} - \ln|u| + C \\
&= \frac{\sec^2 x}{2} - \ln|\sec x| + C
\end{aligned}$$

Let's TRY IT! Find $\int \tan^5 x\,dx$.

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Integration of $\tan^n x$

當要計算 $\int \tan^n x\,dx$ 時，我們會採用以下方法：

➢ ***n* 是雙數 (*n* is even)** ➔ 代換積分法 (Let $u = \tan x$)

e.g.1 Find $\int \tan^2 x\,dx$

Let $u = \tan x$,
then $du = \sec^2 x\,dx$

$$\begin{aligned}\therefore \int \tan^2 x\,dx &= \int (\sec^2 - 1)\,dx \\ &= \int \sec^2 x\,dx - \int 1\,dx \\ &= \int du - \int dx \\ &= u - x + C \\ &= \tan x - x + C\end{aligned}$$

e.g.2 Find $\int \tan^4 x\,dx$

Let $u = \tan x$,
then $du = \sec^2 x\,dx$

$$\begin{aligned}\therefore \int \tan x^4\,dx &= \int \tan^2 x \cdot \tan^2 x\,dx \\ &= \int \tan^2 x\,(\sec^2 x - 1)\,dx \\ &= \int \tan^2 x \sec^2 x\,dx - \int \tan^2 x\,dx \\ &= \int u^2\,du - \int \tan^2 x\,dx \\ &= \frac{u^3}{3} - (\tan x - x) + C \\ &= \frac{\tan^3 x}{3} - \tan x + x + C\end{aligned}$$

註：同學亦可用歸約公式 (Reduction Formula) 求 $\int \tan^n x\,dx$，惟歸約公式不在 M2 考核範圍內。

Let's TRY IT! Find $\int \tan^6 x\,dx$.



Integration of $\sec^n x$

當要計算 $\int \sec^n x\,dx$ 時，我們會採用以下方法：

- ***n* 是單數 (*n* is odd)** ➔ 分部積分法 (Integration by parts) [Ch.9(c)]
- ***n* 是雙數 (*n* is even)** ➔ 代換積分法 (Let $u = \tan x$)

有用恒等式：

$$\sec^2 x = \tan^2 x + 1$$

e.g.1 Find $\int \sec^2 x\,dx$

Let $u = \tan x$,
then $du = \sec^2 x\,dx$

$$\therefore \int \sec^2 x\,dx$$
$$= \int du$$
$$= u + C$$
$$= \tan x + C$$

e.g.2 Find $\int \sec^4 x\,dx$

Let $u = \tan x$,
then $du = \sec^2 x\,dx$

$$\therefore \int \sec^4 x\,dx$$
$$= \int \sec^2 x \cdot \sec^2 x\,dx$$
$$= \int (\tan^2 x + 1) \cdot \sec^2 x\,dx$$
$$= \int (u^2 + 1)\,du$$
$$= \frac{u^3}{3} + u + C$$
$$= \frac{\tan^3 x}{3} + \tan x + C$$



Find $\int \sec^6 x\,dx$.

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Integration of $(\tan^m x \cdot \sec^n x)$

當要計算 $\int \tan^m x \sec^n x \, dx$ 時，我們會根據不同情況採用以下方法：

- ***m* 是單數 (*m* is odd)** ➔ 代換積分法 (Let $u = \sec x$)
- ***n* 是雙數 (*n* is even)** ➔ 代換積分法 (Let $u = \tan x$)
- ***m* 是雙數，*n* 是單數** ➔ 分部積分法 (Integration by parts)

e.g. Find $\int \tan^3 x \sec^5 x \, dx$.

Let $u = \sec x$, then $du = \sec x \tan x \, dx$

$$\begin{aligned}
\int \tan^3 x \sec^5 x \, dx &= \int \tan^2 x \sec^4 x \cdot (\sec x \tan x) \, dx \\
&= \int (\sec^2 x - 1) \sec^4 x \cdot (\sec x \tan x) \, dx \\
&= \int (u^2 - 1) u^4 \, du \\
&= \int (u^6 - u^4) \, du \\
&= \frac{u^7}{7} - \frac{u^5}{5} + C \\
&= \frac{\sec^7 x}{7} - \frac{\sec^5 x}{5} + C
\end{aligned}$$



[A.Math 1984 Paper II Q.5 Modified]

Find $\int \tan^2 \theta \sec^2 \theta \, d\theta$. ***(4 marks)***

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Integration by Trigonometric Substitution

若要計算如 $\int \sqrt{a^2 \pm x^2}\, dx$ 及 $\int \frac{1}{\sqrt{a^2 \pm x^2}}\, dx$ 此類型的積分，建議可試用三角代換積分法 **(Integration by trigonometric substitution)** :

算式	代入之三角函數	所用之恒等式
$\sqrt{a^2 - x^2}$	Let $x = a\sin\theta$ or $a\cos\theta$ $(-\frac{\pi}{2} \le \theta \le \frac{\pi}{2})$	$1 - \sin^2\theta = \cos^2\theta$
$\sqrt{a^2 + x^2}$	Let $x = a\tan\theta$ $(-\frac{\pi}{2} \le \theta \le \frac{\pi}{2})$	$1 + \tan^2\theta = \sec^2\theta$
$\sqrt{x^2 - a^2}$	Let $x = a\sec\theta$ $(0 \le \theta < \frac{\pi}{2}$ or $\pi \le \theta < \frac{3\pi}{2})$	$\sec^2\theta - 1 = \tan^2\theta$

e.g.1 Find $\int \frac{1}{\sqrt{9 - x^2}}\, dx$

Let $x = 3\sin\theta$,
then $dx = 3\cos\theta\, d\theta$

$$\therefore \int \frac{1}{\sqrt{9 - x^2}}\, dx$$
$$= \int \frac{1}{\sqrt{9 - 9\sin^2\theta}} \cdot 3\cos\theta\, d\theta$$
$$= \int \frac{1}{\sqrt{9\cos^2\theta}} \cdot 3\cos\theta\, d\theta$$
$$= \int \frac{1}{3\cos\theta} \cdot 3\cos\theta\, d\theta$$
$$= \int 1\, d\theta$$
$$= \theta + C$$
$$= \sin^{-1} x + C$$

e.g.2 Find $\int \frac{1}{\sqrt{9 - 4x^2}}\, dx$

Let $x = \frac{3}{2}\sin\theta$,

then $dx = \frac{3}{2}\cos\theta\, d\theta$

$$\therefore \int \frac{1}{\sqrt{9 - 4x^2}}\, dx$$
$$= \int \frac{1}{2\sqrt{\frac{9}{4} - x^2}}\, dx$$
$$= \int \frac{1}{2\sqrt{\frac{9}{4} - \frac{9}{4}\sin^2\theta}} \frac{3}{2}\cos\theta d\theta$$
$$= \int \frac{1}{2\left(\frac{3}{2}\cos\theta\right)} \frac{3}{2}\cos\theta d\theta$$
$$= \int \frac{1}{2}\, d\theta$$
$$= \frac{\theta}{2} + C$$
$$= \frac{1}{2}\sin^{-1}\frac{2x}{3} + C$$

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Integration by Trigonometric Substitution

[A.Math 2002 Q.4 Modified]

Find $\int \frac{1}{\sqrt{1-x^2}}\,dx$. [Hint: Let $x = \cos\theta$.] ***(4 marks)***

[A.Math 1994 Paper II Q.10(a) Modified]

Using the substitution $x = \tan\theta$, find $\int \frac{dx}{1+x^2}$. ***(4 marks)***

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Integration by Trigonometric Substitution

[A.Math 1986 Paper II Q.11(a) Modified]

Using the substitution $x = 2\sin\theta$, find $\int \sqrt{4-x^2}\,dx$. ***(6 marks)***

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HKCEE Questions (level 1)

1. ***[A.Math 1994 Paper II Q.1]***

Find $\int (\sin x - \cos x)^2 \, dx$. ***(4 marks)***

2. ***[A.Math 1981 Paper II Q.1]***

Find $\int (1 + \cos\theta)^2 \, d\theta$. ***(5 marks)***

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HKCEE Questions (Level 1)

3. ***[A.Math 1983 Paper II Q.2]***

Use the substitution $u = x^2$ to find $\int x \sin^2(x^2)\, dx$. ***(5 marks)***

4. ***[A.Math 1989 Paper II Q.4]***

(a) Find $\int \cos^2 2x\, dx$.

(b) Using the result in (a), find $\int \sin^2 2x\, dx$. ***(5 marks)***

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HKCEE Questions (Level 2)

5. ***[A.Math 1996 Paper II Q.9(a) Modified]***

Find $\int \sin^5 x \, dx$. [Hint Let $t = \cos x$.] ***(4 marks)***

6. ***[A.Math 1998 Paper II Q.7 Modified]***

Using the substitution $u = \sin\theta$, find $\int \cos^5\theta \sin^2\theta \, d\theta$. ***(6 marks)***

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HKCEE Questions (Level 2)

7. ***[A.Math 1985 Paper II Q.8(a) Modified]***

Find $\int \sin^3 t \cos^4 t \, dt$. ***(4 marks)***

8. ***[A.Math 1997 Paper II Q.11(a)]***

Using the substitution $u = \cot\theta$, find $\int \cot^n \theta \csc^2 \theta \, d\theta$, where n is a non-negative integer. ***(3 marks)***

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HKCEE Questions (Level 2)

9. ***[A.Math 1987 Paper II Q.8(a)]***

Using the substitution $u = \tan x$, find $\int \tan^{n-2} x \sec^2 x \, dx$, where n is a non-negative integer and $n \geq 2$. ***(4 marks)***

10. ***[A.Math 1990 Paper II Q.3]***

Using the substitution $u = \sin^2 x$, find $\int \frac{\sin x \cos x}{\sqrt{9\sin^2 x + 4\cos^2 x}} dx$. ***(5 marks)***

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HKCEE Questions (Level 2)

11. ***[A.Math 1986 Paper II Q.7]***

Let $y = \frac{\tan^3 \theta}{3} - \tan\theta$. Find $\frac{dy}{dx}$ in terms of $\tan\theta$.

Hence, or otherwise, find $\int \tan^4 \theta \, d\theta$. ***(6 marks)***

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HKCEE Questions (Level 2)

12. ***[A.Math 1987 Paper II Q.4 Modified]***

Using the substitution $x = \sin\theta$, find $\int \frac{2x^2}{\sqrt{1-x^2}}\,dx$. ***(6 marks)***

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HKDSE Questions (level 2)

13. ***[M2 2014 Q.5]***

(a) Find $\int \frac{dx}{\sqrt{9-x}}$, where $x < 9$.

(b) Using integration by substitution, find $\int \frac{dx}{\sqrt{9-x^2}}$, where $-3 < x < 3$.

(6 marks)

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HKCEE Questions (Level 3)

14. ***[A.Math 2000 Paper II Q.8(a)(b)]***

(a) Find $\int \cos 3x \cos x \, dx$ ***(3 marks)***

(b) Show that $\dfrac{\sin 5x - \sin x}{\sin x} = 4\cos 3x \cos x$.

Hence, or otherwise, find $\int \dfrac{\sin 5x}{\sin x} \, dx$. ***(4 marks)***

HKCEE Questions (Level 3)

15. ***[A.Math 1984 Paper II Q.7(b)]***

(i) Use the substitution $u = \sin\phi$, find $\int \frac{\cos\phi}{\sin^4\phi}\, d\phi$.

(ii) Use the substitution $u = \tan\phi$ and the result of (i), find $\int \frac{3\sqrt{1+x^2}}{x^4}\, dx$.

(13 marks)

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HKCEE Questions (Level 3)

16. ***[A.Math 1998 Paper II Q.4]***

The slope at any point (x, y) of a curve is given by $\frac{dy}{dx} = \cos^2 x$. If the curve passes through the point $(\frac{\pi}{2}, \pi)$, find its equation. ***(5 marks)***

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HKCEE Questions (Level 3)

17. ***[A.Math 2006 Q.10]***

The slope at any point (x, y) of a curve is given by $\frac{dy}{dx} = 3 + 2\cos 2x$. If the curve passes through the point $(\frac{\pi}{4}, \frac{3\pi}{4})$, find its equation. ***(5 marks)***

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HKCEE Questions (Level 3)

18. ***[A.Math 1984 Paper II Q.3]***

The slope at any point (x, y) of a curve is given by $\frac{dy}{dx} = 4\sin^2 x + 1$. If the curve passes through the x-axis at $x = \frac{\pi}{2}$, find the equation of the curve. ***(6 marks)***

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HKCEE Questions (Level 3)

19. ***[A.Math 1996 Paper II Q.6]***

The slope at any point (x, y) of a curve is given by $\frac{dy}{dx} = \tan^3 x \sec x$. If the curve passes through the origin, find its equation. [Hint Let $u = \sec x$.] ***(6 marks)***

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